DGMS JSO GEOLOGY 2019 • ISOTOPE GEOCHEMISTRY & GEOCHRONOLOGY

Question 97: Initial $^{87}Sr/^{86}Sr$ Ratio of the Earth & BABI

Radioactive Isochron Equation, Basaltic Achondrites, and Primordial Planetary Standards

Q.97: The initial $^{87}Sr/^{86}Sr$ ratio of the earth has been estimated from which one of the following?

(a) Granite

(b) Kimberlite

(c) Chondrite

✓ **(d) Achondrite**

Correct Option: (d) Achondrite

1. Isotopic Principle: The Rb-Sr Decay System

Rubidium-87 (β^- emitter with decay constant $\lambda = 1.42 \times 10^{-11}$ yr^{-1}, half-life 48.8 Gyr) decays to radiogenic Strontium-87. The isotopic evolution of strontium in any closed geological reservoir is described by the isochron equation:

$$\left(^{87}Sr / ^{86}Sr\right)_t = \left(^{87}Sr / ^{86}Sr\right)_0 + \left(^{87}Rb / ^{86}Sr\right) \times \left(e^{\lambda t} - 1\right)$$

where:

- $(^{87}Sr / ^{86}Sr)_t$ = Present-day measured ratio
- $(^{87}Sr / ^{86}Sr)_0$ = Initial ratio at time of planetary differentiation (4.56 Ga)
- $(^{87}Rb / ^{86}Sr)$ = Chemical parent-to-stable-daughter ratio

Because radioactive decay continuously increases the $^{87}Sr/^{86}Sr$ ratio over geological time in any reservoir containing rubidium, determining the pristine initial ratio of the proto-Earth at 4.56 Ga requires finding an ancient planetary material that formed with essentially **zero rubidium (Rb ≈ 0)**.

2. Why Achondrites? The Discovery of BABI

Basaltic achondrites (a group of differentiated stony meteorites comprising eucrites, howardites, diogenites, and angrites) formed by igneous melting on differentiated planetesimals (such as the asteroid 4 Vesta) at 4.56 Ga:

- **Extreme Volatile Depletion:** During planetary differentiation, volatile alkali elements like Rubidium (Rb) were heavily volatilized and lost, while refractory alkaline earth elements like Strontium (Sr) were concentrated into plagioclase-rich basaltic melts.
- **Virtually Zero Radiogenic Ingrowth:** In basaltic achondrites, the Rb/Sr ratio is so vanishingly small (Rb/Sr < 0.005) that negligible radiogenic ^{87}Sr has accumulated since the formation of the Solar System.
- **BABI (Basaltic Achondrite Best Initial):** In 1969, D. A. Papanastassiou and G. J. Wasserburg analyzed basaltic achondrites (e.g., Angra dos Reis, Moore County, Juvinas) and established the classic reference value:

$$\mathbf{(^{87}Sr / ^{86}Sr)_{BABI} = 0.69897 \pm 0.00003}$$

This BABI value represents the canonical baseline initial $^{87}Sr/^{86}Sr$ ratio of the Bulk Silicate Earth (BSE), Moon, and inner solar system rocky bodies.

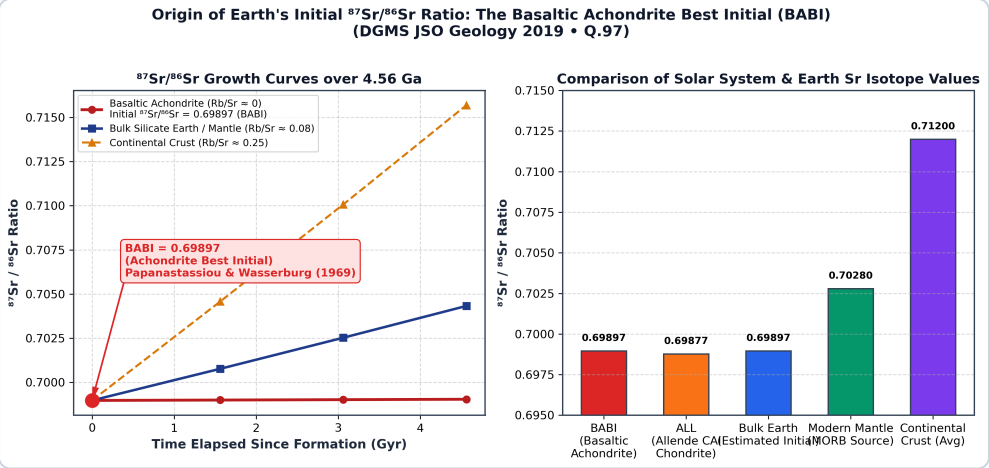


Figure 97.1: Left: Growth curves of $^{87}Sr/^{86}Sr$ for Basaltic Achondrites (Rb/Sr ≈ 0), Bulk Earth Mantle, and Continental Crust. Right: Comparison of benchmark Solar System initial Sr standards highlighting BABI (Basaltic Achondrite Best Initial = 0.69897). Based on White, Geochemistry (Chapter 9) and Faure, Principles of Isotope Geology.

3. Systematic Option Evaluation

Option	Rock / Meteorite Class	Typical $^{87}Sr/^{86}Sr$ & Characteristics	Validity as Primordial Earth Initial Reference
(a)	Granite	0.708 – 0.730+ (High Rb/Sr, highly radiogenic crustal melt)	**Incorrect.** Granites are highly differentiated felsic rocks enriched in Rb; they have elevated, variable $^{87}Sr/^{86}Sr$ due to billions of years of crustal radiogenic decay.
(b)	Kimberlite	0.703 – 0.706 (Deep subcontinental mantle source)	**Incorrect.** Kimberlites are Phanerozoic or Proterozoic mantle-derived pipes that have experienced billions of years of mantle evolution; they do not preserve the primordial 4.56 Ga initial ratio.
(c)	Chondrite		**Incorrect.** While undifferentiated, whole-rock

Option	Rock / Meteorite Class	Typical $^{87}Sr/^{86}Sr$ & Characteristics	Validity as Primordial Earth Initial Reference
		0.700 – 0.730 (Whole rock, Rb/Sr ≈ 0.08)	chondrites contain moderate volatile Rb and have evolved significant radiogenic ^{87}Sr over 4.56 Ga. (Only refractory calcium-aluminium inclusions (CAIs) like Allende ALL = 0.69877 have low Rb/Sr, but the classic terrestrial reference standard is BABI).
(d)	**Achondrite**	**0.69897 ± 0.00003 (BABI)**	**Correct.** Basaltic achondrites provide the universal reference standard (BABI) for the initial $^{87}Sr/^{86}Sr$ of the Earth and inner planets due to volatile depletion of Rb and ancient 4.56 Ga crystallization.

> **Key Geochemical Takeaway**
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> The **BABI** (Basaltic Achondrite Best Initial = 0.69897) determined from **basaltic achondrite meteorites** is the universal reference starting point for terrestrial strontium isotope geochemistry. Any terrestrial rock with $^{87}Sr/^{86}Sr > 0.69897$ reflects the cumulative decay of ^{87}Rb over planetary history.

4. Standard Textbook References

- **William M. White**, *Geochemistry* (2nd Edition, Wiley-Blackwell, 2020), Chapter 9: "Radiogenic Isotope Geochemistry", Section 9.2: The Rb-Sr System and BABI, pp. 360–375.

- **Gunter Faure & Teresa M. Mensing**, *Isotopes: Principles and Applications* (3rd Edition, Wiley, 2005), Chapter 4: "The Rubidium-Strontium Method", Section 4.5: Initial $^{87}Sr/^{86}Sr$ Ratios of Meteorites and the Earth, pp. 68–85.
- **Alan P. Dickin**, *Radiogenic Isotope Geology* (3rd Edition, Cambridge University Press, 2018), Chapter 3: "The Rb-Sr Method", pp. 55–78.
- **Papanastassiou, D. A., & Wasserburg, G. J.** (1969), *Initial Strontium Isotopic Abundances and the Resolution of Small Time Differences in the Formation of Planetary Bodies*, Earth and Planetary Science Letters, 5, pp. 361–376.